KenganSocial pricing model

Credits, usage limits and prices per plan, worked out from what each service costs per second and per execution. 23 October 2026. Generated from `scripts/pricing-model.mjs`; re-run it when a provider changes its prices.

> **Result.** With the new allowances every paid plan keeps a gross margin of **at least 60%** even when a customer uses 100% of every allowance, on the 10%-discounted annual price, after VAT and Payfast fees, with a 20% safety buffer on every third-party price. Prices stay the same (R299, R699, R3,499, R8,999). An automated test (`server/tests/security/pricing.test.ts`) fails if the app's limits drift from this model or a plan drops below 60%.

Plan	Price	Cost at 100% use	Margin monthly	Margin annual	Lowest price for 60%	Margin before (annual)
Creator Starter	R299	R69,45	68.2%	65.1%	R256	52.2%
Creator Pro	R699	R164,81	68.3%	65.2%	R597	23.0%
Agency	R3 499	R847,81	67.8%	64.7%	R3 037	9.6%
Enterprise	R8 999	R847,81	84.9%	83.7%	R3 037	—

"Lowest price for 60%" is the list price at which the annual price still reaches 60% at full use. "Before" costs the old allowances with the same unit costs. Enterprise runs on the Agency allowances in the app; extras are quoted from the unit costs below.

1. How the costs are measured

Every action is costed three ways and the plan limits are counted in the same units:

- **By the second:** video work (transcription, analysis, rendering, exporting) is measured in *VM-seconds*: seconds of one 4-CPU server (e2-standard-4, Johannesburg). The server costs ~US$140/month; capacity is bought for peaks, so only 50.0% of every second is counted as useful. One useful VM-second costs **R0,00189**.
- **By execution:** each call to an outside AI or media API is costed from the provider's price, at its worst case (largest output it is allowed to produce, most expensive model in the fallback chain).
- **By plan limits:** a plan's cost is every allowance used in full, times the unit's worst-case cost.

```
rand cost of an API call = US$ price × R17.5 (exchange) × 1.15 (VAT on imported services) × 1.2 (safety buffer)
net revenue           = list price ÷ 1.15 (treated as VAT-inclusive)
payment fee           = (price × 3.2% + R2.00) × 1.15
gross margin          = (net revenue − cost at 100% use − payment fee) ÷ net revenue   ≥ 60%
```

Compute per second of media (e2-standard-4, VM-seconds per second)

Work	VM-s per media second	Cost per media minute	Basis
Transcription, Whisper base (Creator Starter)	0.1	R0,011	faster-whisper int8 on CPU
Transcription, Whisper small (Creator Pro)	0.3	R0,034	≈3× base
Transcription, Whisper medium (Agency, trial)	0.9	R0,102	≈9× base
Scene and face analysis of the source	0.05	R0,006	PySceneDetect, YuNet
Clip render 1080×1920 with face tracking	1.5	R0,170	FFmpeg libx264 veryfast
Editor export	1.5	R0,170	same encoder
Remotion render (launch, tutorial, video ad)	9	R1,021	20-s template took 1-3 min (DEPLOY.md); launch videos 90-274 s

Third-party prices used (US$, before buffer)

Service	Price	Used by
Claude Haiku 4.5 (OpenRouter)	$1 / $5 per M tokens	all text AI, clip finding, storyboards
Claude Opus (lyrics structuring)	$5 / $25 per M tokens	lyrics analyser
Fast AI image (Nano Banana 2)	$0.0672 each	carousels, thumbnails, AI Writer images
Best AI image (GPT Image, medium, up to 1536 px)	$0.08 each	carousels
ElevenLabs Multilingual v2	$0.10 per 1,000 characters	video ad voiceover
Treblo music	$0.1 per 30-s track (estimate)	video ads, launch videos
Perplexity Sonar	$0.007 per query	trending
OpenAI whisper-1	$0.006 per minute	lyrics only
Remotion Automators licence	$0.01 per render, $100/month minimum	all Remotion renders

2. Cost of each unit a plan counts

Unit	Worst-case cost	What it covers
AI credit	R0,36	up to US$0.015 of model use: a Haiku call with 4k tokens in and 1.5k out costs US$0.0115
Scheduled post	R0,15	platform downloads ~50 MB once, plus the publish job
Export unit (5 min of video)	R0,85	300 s × 1.5 VM-s
Render unit	R4,44	the dearest of: a 30-s launch video (capture, storyboard with screenshots, music, Remotion), a 25-s video ad (voice, music, Remotion), or 5 min of AI edit
Transcription minute	R0,011 / R0,034 / R0,102	base / small / medium Whisper
AI clip project	R4,21 (30 min) · R5,56 (60 min) · R9,41 (120 min)	source analysis + one 8.5k-token chunk per ~20 min of speech (max 8k tokens out) + 10 clips of 60 s rendered; transcription is counted separately
Storage, per GB a month	R3,07	stored and downloaded once

AI credits per action

Action	Credits	Worst-case provider cost	Change
Text generation (hook, script, repurpose, AI Writer, brand voice, DM draft)	1	≤ $0.015	DM drafts were free; AI Writer used a separate 50-credit balance any plan got
Captions	1 per platform	≤ $0.015 each	was 1 for all platforms together
Thumbnail headline ideas	1	≤ $0.015	was free (without credits: built from the clip's words)
Trending refresh (manual)	2	~$0.019	was free; refreshing every 5 min could cost ~R3,000/month per workspace
Brand set-up from a website	5	~$0.07	was free on any plan with credits
AI Edit: learn a style	5	~$0.045	unchanged
Fast AI image	5	$0.0672	was 2 (1 for a thumbnail background)
Best AI image	6	$0.08	unchanged
Lyrics analysis (6-min song)	17	~$0.25	was free on every plan, including Free

Counted by length (per second, rounded up)

- **Transcription** is counted in minutes (was whole hours: a 61-minute video used 2 hours, and the standalone transcribe used 1 hour whatever its length).
- **Editor exports** count 1 per started 5 minutes of exported video (was 1 per export of any length).
- **AI edits** count 1 render per started 5 minutes of source (was 1 for up to 3 hours of video).
- **Longest clip source**: Creator Starter 30 min, Creator Pro 60 min (was 90), Agency 120 min (was 180).

3. Plans: allowances and full-use cost

Allowance per month	Creator Starter before → now	Creator Pro before → now	Agency before → now
Scheduled posts	30	100	Unlimited → **1 000**
Editor exports (5-min units)	10 → **8**	30 → **20**	150 → **80**
Transcription (minutes)	180 → **120**	600 → **420**	2 400 → **1 200**
AI clip projects	3	12 → **7**	40 → **15**
AI credits	100 → **60**	250 → **100**	1 300 → **500**
Video renders	6 → **3**	25 → **5**	150 → **25**
Storage (GB)	2	10 → **5**	50 → **20**
Longest clip source	30 min	90 → **60 min**	180 → **120 min**

Unchanged: connected accounts (2 / 3 / 6 / 25), DM automation rules (0 / 10 / 50 / unlimited; rules cost nothing to run, AI drafts are paid in credits), client approvals (Creator Pro 5 clients, Agency unlimited). Agency "unlimited posts" becomes 1,000 a month (33 a day).

Creator Starter: R299/month (R269/month billed yearly)

Cost line at 100% use	Posts	Exports	Transcription	AI clip projects	AI credits	Renders	Storage	Platform (Firestore, sync)	Total
Rand	R4,46	R6,81	R1,36	R12,64	R21,74	R13,32	R6,13	R3,00	**R69,45**

Billing	Price	Net of VAT	Payfast fee	Cost incl. fee	Gross profit	Margin
monthly	R299	R260,00	R13,30	R82,76	R177,24	68.2%
annual	R269	R233,91	R12,20	R81,65	R152,26	65.1%

Creator Pro: R699/month (R629/month billed yearly)

Cost line at 100% use	Posts	Exports	Transcription	AI clip projects	AI credits	Renders	Storage	Platform (Firestore, sync)	Total
Rand	R14,87	R17,01	R14,29	R38,89	R36,23	R22,20	R15,33	R6,00	**R164,81**

Billing	Price	Net of VAT	Payfast fee	Cost incl. fee	Gross profit	Margin
monthly	R699	R607,83	R28,02	R192,84	R414,99	68.3%
annual	R629	R546,96	R25,45	R190,26	R356,69	65.2%

Agency: R3 499/month (R3 149/month billed yearly)

Cost line at 100% use	Posts	Exports	Transcription	AI clip projects	AI credits	Renders	Storage	Platform (Firestore, sync)	Total
Rand	R148,68	R68,06	R122,50	R141,14	R181,13	R110,99	R61,32	R14,00	**R847,81**

Billing	Price	Net of VAT	Payfast fee	Cost incl. fee	Gross profit	Margin
monthly	R3 499	R3 042,61	R131,06	R978,88	R2 063,73	67.8%
annual	R3 149	R2 738,26	R118,18	R966,00	R1 772,26	64.7%

Enterprise: R8 999/month (R8 099/month billed yearly)

Cost line at 100% use	Posts	Exports	Transcription	AI clip projects	AI credits	Renders	Storage	Platform (Firestore, sync)	Total
Rand	R148,68	R68,06	R122,50	R141,14	R181,13	R110,99	R61,32	R14,00	**R847,81**

Billing	Price	Net of VAT	Payfast fee	Cost incl. fee	Gross profit	Margin
monthly	R8 999	R7 825,22	R333,46	R1 181,28	R6 643,94	84.9%
annual	R8 099	R7 042,61	R300,34	R1 148,16	R5 894,45	83.7%

4. Free plan, trial and free starter clip

Who	Cost per user at full use	Notes
Free plan	R6,57 / month	no AI, 3 exports, 10 posts, 0.5 GB
7-day trial (now)	R106,93 once	every Agency feature unlocked, Creator Starter quantities (enforced in planGate)
7-day trial (before)	R2 355,99 once	full Agency allowances, no card: a fake sign-up could cost about this much
Free starter clip	~R3,02 each	one per email, 2 per network a day; daily cap now 25 (was 100): at most ~R2 267 a month

5. Fixed costs and break-even

Fixed monthly cost	Rand
Remotion licence minimum ($100, render fees count towards it)	R2 012
Backups, snapshots, domain, email, monitoring	R600
Total	**R2 613**

The server itself is costed per VM-second inside each plan (at 50.0% utilisation); when usage grows, a second server is added and the per-second cost stays the same. Even if every customer used 100% of every allowance, the average paid plan contributes R2 325 a month, so the fixed costs are covered from about **15 paying customers** on the cheapest plan (2 on an even mix). Typical customers use far less than 100%, so real break-even comes sooner.

6. Sensitivity

Scenario	Creator Starter annual margin	Creator Pro annual margin	Agency annual margin	Enterprise annual margin
Base case (R17.5/$, 20% buffer, prices VAT-inclusive)	65.1%	65.2%	64.7%	83.7%
No safety buffer (prices exactly as published)	67.2%	67.5%	67.2%	84.7%
Weaker rand: R20 per US$	63.3%	63.3%	62.6%	82.9%
Not VAT-registered (no VAT taken from the price)	69.6%	69.8%	69.3%	85.8%

With the 20% buffer kept on, every plan holds 60% up to **R23.06 per US$** (equivalent to about R28 if providers charge exactly their published prices). If the rand weakens past that, change `usdZar` in the model and re-run: the pricing test then fails and points at the plan that needs a smaller allowance or a higher price.

7. What changed in the app

- **Limits** (server and client `planLimits.ts`, pricing page, Billing): new allowances above; transcription counted in minutes.
- **Per-second counting:** transcription by the minute everywhere (clips, repurpose, AI edit, standalone transcribe); editor exports and AI edits per 5 minutes of video.
- **Per-execution counting:** captions per platform; AI images 5 or 6 credits everywhere (carousels, thumbnails, AI Writer); trending refresh, website set-up, thumbnail headlines, DM drafts and lyrics now cost credits, with refunds when the work fails or is served from cache.
- **AI Writer** uses the plan's monthly AI credits; the old never-resetting 50-credit balance is gone. Its legacy video render uses the render allowance.
- **Trial** keeps every Agency feature but with Creator Starter quantities.
- **Free starter clip** daily cap 25 by default (`FREE_CLIP_DAILY_CAP`).
- **Guard rail:** `server/tests/security/pricing.test.ts` checks the app's limits, prices and credit costs against this model and fails if any paid plan falls below 60%.

8. Findings still open

- **Prices to confirm with the providers:** Treblo's per-track price is not published (estimated at $0.1); the lyrics model is costed at Opus 4.5/4.6 rates; Payfast recurring-card fees; the VM's exact Johannesburg price. Update `scripts/pricing-model.mjs` and re-run.
- **Measure on the real server:** the VM-seconds per media second come from local runs and the deploy notes. Time a 10-minute clip project and a 25-s video ad on the VM and update `COMPUTE`.
- **Two legacy endpoints are still unmetered:** `/api/ai/video/analyze` (Whisper base on uploads up to 100 MB) and `/api/ai/video/edit`. Their size cap limits the exposure; meter or retire them next.
- **Advertised but not enforced:** the connected-accounts limit (2/3/6/25) and storage (only enforced for the Agency library).
- **Fallbacks are cheaper, not dearer:** when a cheaper provider answers (Gemini Flash, Pollinations' free images), real costs are below this model. The worst case is what the margin is guaranteed on.

Sources

- Claude Haiku 4.5 on OpenRouter: $1 / $5 per million tokens: https://openrouter.ai/anthropic/claude-haiku-4-5/pricing
- Gemini API pricing (Flash, Flash-Lite, Flash Image): https://cloud.google.com/gemini-enterprise-agent-platform/generative-ai/pricing
- OpenRouter image models compared (Nano Banana 2 $0.0672, GPT Image): https://openrouter.ai/blog/insights/image-generation-models-compared/
- OpenAI image API pricing (GPT Image medium ~$0.053): https://www.modellix.ai/blog/openai-image-generation-api-pricing
- ElevenLabs API pricing ($0.10 per 1,000 characters, Multilingual): https://developer.puter.com/tutorials/elevenlabs-api-pricing/
- Perplexity Sonar API pricing ($1/$1 per M tokens + $5 per 1,000 requests): https://developer.puter.com/tutorials/perplexity-api-pricing/
- Treblo API (credit-based, from ~$11/month; per-track price not published): https://kompozy.io/reviews/treblo
- Remotion licence (Automators: $0.01 per render, $100/month minimum): https://www.remotion.dev/docs/license/pricing
- Google Video Intelligence pricing (not used by the app today): https://cloud.google.com/video-intelligence/pricing
- e2-standard-4 in africa-south1 (~$128/month on demand): https://gcloud-compute.com/e2-standard-4.html
- Firebase / Cloud Storage pricing: https://firebase.google.com/pricing
- Payfast fees (card 3.2% + R2.00): https://payfast.io/fees
- USD/ZAR rate (16.38 on 16 Sep 2026; Oct forecast 16.25-17.33): https://longforecast.com/dollar-to-rand-forecast-2017-2018-2019-2020-2021-usd-to-zar

Third-party price pages disagree in places; where they did, the higher figure was used and a 20% buffer added on top.